

Modul I

2

a) $u = xy + yz + zx$

$x = t \cos t, y = t \sin t, z = t$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

$$\frac{\partial u}{\partial x} = y + 0 + z \Rightarrow y + z$$

$$\frac{dx}{dt} = \cos t - t \sin t$$

$$\frac{\partial u}{\partial y} = x + z + 0 \Rightarrow x + z$$

$$\frac{dy}{dt} = \sin t + t \cos t$$

$$\frac{\partial u}{\partial z} = 0 + y + x \Rightarrow y + x$$

$$\frac{dz}{dt} = 1$$

$$\frac{du}{dt} = (t \sin t + t)(\cos t - t \sin t) + (t \cos t + t)(\sin t + t \cos t) + (t \sin t + t \cos t)(1)$$

$$= t \sin t \cos t - t^2 \sin^2 t + \cos^2 t - \sin^2 t + t \cos t \sin t + t^2 \cos^2 t + \sin^2 t + \cos^2 t + t \sin t + t \cos t$$

$$= \frac{\pi}{4} \left(\frac{1}{\sqrt{2}} \right) + \frac{\pi}{4} \left(\frac{1}{\sqrt{2}} \right)$$

$$= t(\sin t + 1)(\cos t - \sin t) + t(\cos t + 1)(\sin t + \cos t) + t(\sin t + \cos t)$$

$$= \frac{\pi}{4} \left(\frac{1}{\sqrt{2}} + 1 \right) \left(\frac{\pi}{4\sqrt{2}} - \frac{1}{\sqrt{2}} \right) + \frac{\pi}{4} \left(\frac{1}{\sqrt{2}} + 1 \right) \left(\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} \right) + \frac{\pi}{4} \left(\frac{1}{\sqrt{2}} + \frac{\pi}{4\sqrt{2}} \right)$$

6 $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$

$\frac{\partial(u, v, w)}{\partial(x, y, z)} \Rightarrow$

$\frac{\partial u}{\partial x}$	$\frac{\partial v}{\partial x}$	$\frac{\partial w}{\partial x}$
$\frac{\partial u}{\partial y}$	$\frac{\partial v}{\partial y}$	$\frac{\partial w}{\partial y}$
$\frac{\partial u}{\partial z}$	$\frac{\partial v}{\partial z}$	$\frac{\partial w}{\partial z}$

$\Rightarrow$

$\frac{\partial u}{\partial x}$	$\frac{\partial u}{\partial y}$	$\frac{\partial u}{\partial z}$
$\frac{\partial v}{\partial x}$	$\frac{\partial v}{\partial y}$	$\frac{\partial v}{\partial z}$
$\frac{\partial w}{\partial x}$	$\frac{\partial w}{\partial y}$	$\frac{\partial w}{\partial z}$

6 $\Rightarrow$

$2x$	$2y$	$2z$
$y+z$	$x+z$	$y+x$
1	1	1

$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 2x(x+z - y+x) - 2y(y+z - x+y) + 2z(y+z - x+x)$
 $= 2x(2x+z-y) - 2y(2y+z-x) + 2z(2z+y-x)$
 $= 2x(2x+z-y) - 2y(2y+z-x) + 2z(2z+y-x)$
 $= 2x(2x+z-y) - 2y(2y+z-x) + 2z(2z+y-x)$
 $= 0$

c $f(x,y) = x^3 + y^3 - 3x - 12y + 20$

$f_x(x,y) = 3x^2 + 0 - 3 - 0 + 0$

$f_x = 3x^2 - 3$

$f_{xx} = 6x - 0$

$f_y = 0 + 3y^2 - 0 - 12 + 0 = 3y^2 - 12$

$f_{yy} = 0 + 6y - 0$

$f_x = 3x^2 - 3$

$f_{xy} = 0$

$3x^2 - 3 = 0$

$x^2 - 1 = 0$

$x = \pm 1$

$3y^2 - 12 = 0$

$y^2 - 4 = 0$

$y = \pm 2$

The stationary points are

$(1, 2), (1, -2), (-1, 2), (-1, -2)$

	1, 2	1, -2	-1, 2	-1, -2
A	6	6	-6	-6
B	0	0	0	0
C	12	-12	12	-12
$AC - B^2$	+72	-72	-72	+72
Conclusion	min. pt	saddle	saddle	max. pt

* Continued after Module I and Module II (5)

find $\Rightarrow$ max. value
 min. value

continued in
 last page

P.T.O

Module II

4

b $y(2x-y+1)dx + x(3x-4y+4)dy = 0$

$M = y(2x-y+1)$

$M = 2yx - y^2 + y$

$\frac{\partial M}{\partial y} = 2x - 2y + 1$

$\frac{\partial}{\partial y}$

$N = x(3x-4y+4)$

$N = 3x^2 - 4xy + 4x$

$\frac{\partial N}{\partial x} = 6x - 4y + 4$

$\frac{\partial}{\partial x}$

$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2x - 6x - 2y - 4y + 1 + 4$

$= -4x - 2y + 5$

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Grac

$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2x - 2y + 1 - (6x - 4y + 4)$

$= -4x - 2y + 3$

It is closer to M

Im

$\frac{1}{n} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{-4x - 2y + 3}{y(2x-y+1)}$

$= \frac{-2 + 2 - 3}{y}$

$= \frac{-3}{y} = -3 \int \frac{1}{y} dy = -3 \ln y$

$$IF = e^{-\int p(x) dx} = e^{-\int -3/y dx} = e^{\log y^3} = y^3$$

multiply by y^3

$$y^{-2}(2x - y + 1) dy + xy^3(3x - 4y + 4)$$

$$C (D^2 + 2D + 1)y = x^2 + 2x$$

$D \rightarrow m$

$$A.F.: m^2 + 2m + 1 = 0$$

$$m = -1, -1$$

C.F.:

$$y_c = (C_1 + C_2 x) e^{-x}$$

$$P.E = y_p = \frac{1}{D^2 + 2D + 1}$$

$$= \frac{1}{(1+D)^2} (2x + x^2)$$

$$= (1+D)^{-2} (2x + x^2)$$

$$(1+D)^{-2} = 1 - 2D + 3D^2 - 4D^3 + \dots$$

$$= 2x + x^2 - 4D(2x + x^2) + 6D^2(x^2) + 3D^2(2x)$$

$$= 2x + x^2 - 4 \cdot 1 - 2 \cdot 2x + 6(1) + 3 \cdot 2$$

$$y_p = 2x + x^2 - 4 - 4x + 6 = x^2 - 2x + 2$$

$$y_p = x^2 - 2x + 2$$

Binomial series.
 $1 - x + x^2 - x^3 + x^4 - \dots$

Module I and Module II

5

a $u = z + 3y^2 - z^3$
 $v = 4x^2 y^2$

$w = 2z^2 - xy$

$\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at $(1,-1,0)$

Using Jacobians

$\frac{\partial u}{\partial x}$	$\frac{\partial v}{\partial x}$	$\frac{\partial w}{\partial x}$
$\frac{\partial u}{\partial y}$	$\frac{\partial v}{\partial y}$	$\frac{\partial w}{\partial y}$
$\frac{\partial u}{\partial z}$	$\frac{\partial v}{\partial z}$	$\frac{\partial w}{\partial z}$

1	8y	-6y
8x	8y	0
0	-1	-3z

$1(8y \cdot 4z - 0) - 6y(8x \cdot 4z + 0) - 3z(-8x - 8y^2)$

$32yz - 192xz + 24xz + 24y^2z$

$0 - 0 + 0 + 24(1) = 24$

Continuation of 2(c)

The max point is $(-1, -2)$

sub in ①

$$(-1)^3 + (-2)^3 - 3(-1) - 12(-2) + 20 = 38$$

The min point is $(1, 2)$

~~$$(-1)^3 + (-2)^3 - 3(-1) - 12(-2) + 20 = 38$$

$$(1)^3 + (2)^3 - 3(1) - 12(2) + 20 = -2$$~~

max
given
in previous
page